No.

Date

高数B上. HW6.

习题3.3.1. $\int \frac{x-1}{x^2+6x+8}\,dx$

$\frac{x-1}{x^2+6x+8} = \frac{x-1}{(x+2)(x+4)} = \frac{A}{x+2} + \frac{B}{x+4}$. 解得 $A=-\frac{3}{2}$, $B=\frac{5}{2}$

原式 $= -\frac{3}{2}\ln|x+2| + \frac{5}{2}\ln|x+4| + C$.

3. $\int \frac{2x^2-5}{x^4-5x^2+6}\,dx = \int \frac{dx}{x^2-2} + \int \frac{dx}{x^2-3} = \frac{1}{\sqrt{2}}\ln\left|\frac{x-\sqrt{2}}{x+\sqrt{2}}\right| + \frac{1}{2\sqrt{3}}\ln\left|\frac{x-\sqrt{3}}{x+\sqrt{3}}\right| + C$.

7. $\int \frac{dx}{1+x^4}$ 参见第7次习题课讲义最后1题.

9. $\int \frac{e^x dx}{e^{2x}+3e^x+2} = \int \frac{de^x}{(e^x+1)(e^x+2)} = \int \frac{1}{e^x+1} - \frac{1}{e^x+2}\,de^x = \ln\left|\frac{e^x+1}{e^x+2}\right| + C$.

11. $\int \frac{x^3}{x^4+x^2+2}\,dx = \frac{1}{2}\int \frac{x^2\,dx^2}{x^4+x^2+2} = \frac{1}{4}\ln(x^4+x^2+2) - \frac{1}{2\sqrt{7}}\arctan\frac{2x^2+1}{\sqrt{7}} + C$

Rmk. 考前复习 $\int \frac{Bx+C}{x^2+px+q}\,dx = \frac{B}{2}\ln(x^2+px+q) + \frac{C-\frac{Bp}{2}}{\sqrt{q-\frac{p^2}{4}}}\arctan\frac{x+\frac{p}{2}}{\sqrt{q-\frac{p^2}{4}}} + \tilde{C}$

的推导, 或背下来! 这里 $p=1$, $q=2$, $B=1$, $C=0$

16. $\int \sec^4 x\,dx = \int \sec^2 x\,d\tan x = \int (1+\tan^2 x)\,d\tan x = \tan x + \frac{1}{3}\tan^3 x + C$

18. $\int \frac{\cos^3 x}{\sin x+\cos x}\,dx$ 令 $y = x+\frac{\pi}{4}$.

分母 $= \sqrt{2}\sin y$. 分子 $= \left(\frac{\sqrt{2}}{2}\cos y + \frac{\sqrt{2}}{2}\sin y\right)^3 = \frac{\sqrt{2}}{4}(c^3+3c^2s+3cs^2+s^3)$ ($c \triangleq \cos y$, $s \triangleq \sin y$)

原式 $= \frac{1}{4}\int \frac{c^3}{s} + 3c^2 + 3sc + s^2\,dy$

$\int \frac{c^3}{s}\,dy = \int \frac{1-s^2}{s}\,ds = \ln|\sin y| - \frac{1}{2}\sin^2 y + C$

$\int c^2\,dy = \int \frac{1+\cos 2y}{2}\,dy = \frac{y}{2} + \frac{1}{4}\sin 2y + C$

$\int s^2\,dy = \int \frac{1-\cos 2y}{2}\,dy = \frac{y}{2} - \frac{1}{4}\sin 2y + C$

$\int sc\,dy = \int \sin y\,d\sin y = \frac{1}{2}\sin^2 y + C$

$= \frac{1}{4}\ln\left|\sin\left(x+\frac{\pi}{4}\right)\right| + \frac{1}{8}\sin^2\left(x+\frac{\pi}{4}\right) + \frac{x}{2} + \frac{1}{8}\cos 2x + C$

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No.

Date

19. $\int \sin^5 x \cos^2 x\,dx = -\int s^4 c^2\,dc. = -\int c^2(1-c^2)^2\,dc$

$= \int -c^6+2c^4-c^2\,dc = -\frac{1}{7}\cos^7 x + \frac{2}{5}\cos^5 x - \frac{1}{3}\cos^3 x + C.$

22. $\int \frac{dx}{\sin x + 2\cos x} = \frac{1}{\sqrt{5}}\int \frac{d(x+\varphi)}{\sin(x+\varphi)}$ $\tan\varphi = 2.$ $\sin\varphi = \frac{2\sqrt{5}}{5}$, $\cos\varphi = \frac{\sqrt{5}}{5}$

* 记得 $\int \frac{dt}{\sin t} = \int \frac{-d\cos t}{1-\cos^2 t} = \frac{1}{2}\ln\left|\frac{\cos t - 1}{\cos t + 1}\right|.$

原式 $= \frac{1}{2\sqrt{5}}\ln\left|\frac{\cos(x+\varphi)-1}{\cos(x+\varphi)+1}\right| + C$ $\cos(x+\varphi) = \frac{\sqrt{5}}{5}\cos x - \frac{2\sqrt{5}}{5}\sin x.$

$= \frac{1}{2\sqrt{5}}\ln\left|\frac{\cos x - 2\sin x - 1}{\cos x - 2\sin x + 1}\right| + C.$

26. $\int \frac{1-\sqrt{x-1}}{1+\sqrt[3]{x-1}}\,dx$ 令 $t = \sqrt[6]{x-1}.$ $x = t^6+1.$ $dx = 6t^5\,dt.$

$= \int \frac{1-t^3}{1+t^2}\cdot 6t^5\,dt.$ ~~$= 6\int \frac{t^2+t+1}{t+1} t^5\,dt$~~

$= 6\int -t^6+t^4+t^3-t+1+\frac{t}{1+t^2}-\frac{1}{1+t^2}\,dt$ $t = \sqrt[6]{x-1}$

$= -\frac{6}{7}t^7 + \frac{6}{5}t^5 + \frac{3}{2}t^4 - 2t^3 - 3t^2 + 6t + 3\ln|t^2+1| - 6\arctan t + C$

27. $\int \frac{\sqrt{x+1}+\sqrt{x-1}}{\sqrt{x+1}-\sqrt{x-1}}\,dx. = \frac{1}{2}\int (\sqrt{x+1}+\sqrt{x-1})^2\,dx = \int x + \sqrt{x^2-1}\,dx.$

~~$\int \sqrt{x^2-1}\,dx$ 令 $x = \frac{1}{\cos t}.$ $dx = \frac{\sin t}{\cos^2 t}\,dt$~~

~~$\int \frac{\sin^2 t}{\cos^3 t}\,dt = \int \frac{\sin^2 t}{\cos^4 t}\,d\sin t =$~~

$\int \sqrt{x^2-1}\,dx = x\sqrt{x^2-1} - \int x\,d\sqrt{x^2-1}$

$= \int x\frac{x}{\sqrt{x^2-1}}\,dx = \int \sqrt{x^2-1} + \frac{1}{\sqrt{x^2-1}}\,dx$

$\Rightarrow \int \sqrt{x^2-1}\,dx = x\sqrt{x^2-1} - \int \sqrt{x^2-1}\,dx - \int \frac{dx}{\sqrt{x^2-1}}$

$\Rightarrow \int \sqrt{x^2-1}\,dx = \frac{x}{2}\sqrt{x^2-1} - \frac{1}{2}\int \frac{dx}{\sqrt{x^2-1}}$ $\int \frac{dx}{\sqrt{x\pm a^2}} = \ln|x+\sqrt{x^2\pm a^2}| + C$

$= \frac{x}{2}\sqrt{x^2-1} - \frac{1}{2}\ln|x+\sqrt{x^2-1}| + C.$

Campus 原式 $= \frac{x^2}{2} + \frac{x}{2}\sqrt{x^2-1} - \frac{1}{2}\ln|x+\sqrt{x^2-1}| + C.$

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习题3.4.

2. $\int_0^{\ln 2} xe^{-x}dx = -\int_0^{\ln 2} x\,de^{-x} = -xe^{-x}\Big|_0^{\ln 2} + \int_0^{\ln 2} e^{-x}dx$

$= -xe^{-x}\Big|_0^{\ln 2} - e^{-x}\Big|_0^{\ln 2} = -\frac{1}{2}\ln 2 - 0 - (\frac{1}{2} - 1)$

$= \frac{1}{2} - \frac{1}{2}\ln 2$.

5. $\int_0^4 \sqrt{x^2+9}\,dx = \underline{x\sqrt{x^2+9}\Big|_0^4} - \int_0^4 x\,d\sqrt{x^2+9}$

$= 20$

$\int_0^4 x\,d\sqrt{x^2+9} = \int_0^4 \frac{x^2}{\sqrt{x^2+9}}dx = \int_0^4 \sqrt{x^2+9}\,dx - 9\int_0^4 \frac{dx}{\sqrt{x^2+9}}$

$\Rightarrow \int_0^4 \sqrt{x^2+9}\,dx = 20 - \int_0^4 \sqrt{x^2+9}\,dx + 9\int_0^4 \frac{dx}{\sqrt{x^2+9}}$

$\int_0^4 \sqrt{x^2+9}\,dx = 10 + \frac{9}{2}\ln(x+\sqrt{x^2+9})\Big|_0^4 = 10 + \frac{9}{2}\ln 3$.

6. $\int_0^{\frac{1}{2}} \frac{x^2}{\sqrt{1-x^2}}dx = \int_0^{\frac{1}{2}} \frac{x^2-1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}dx = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} - \int_0^{\frac{1}{2}} \sqrt{1-x^2}\,dx$

$\int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^2}} = \arcsin\frac{1}{2} - 0 = \frac{\pi}{6}$.

$\int \sqrt{1-x^2}\,dx \overset{x=\sin\theta}{=\!=} \int \cos^2\theta\,d\theta = \int \frac{1+\cos 2\theta}{2}d\theta = \frac{\theta}{2} + \frac{1}{4}\sin 2\theta$

$= \frac{1}{2}\arcsin x + \frac{x}{2}\sqrt{1-x^2}$.

$\int_0^{\frac{1}{2}} \sqrt{1-x^2}\,dx = \frac{1}{2}\times\frac{\pi}{6} + \frac{1}{4}\cdot\sqrt{\frac{3}{4}} = \frac{\pi}{12} + \frac{\sqrt{3}}{8}$.

$\Rightarrow$ 原式 $= \frac{\pi}{12} - \frac{\sqrt{3}}{8}$.

7. $\int_0^1 \sqrt{4-x^2}\,dx \overset{x=2y}{=\!=} 2\int_0^{\frac{1}{2}} 2\sqrt{1-y^2}\,dy = 4\int_0^{\frac{1}{2}} \sqrt{1-y^2}\,dy = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$.

9. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos x - \cos^3 x}\,dx = 2\int_0^{\frac{\pi}{2}} \sqrt{\cos x}\cdot\sin x\,dx = -2\int_0^{\frac{\pi}{2}} \sqrt{\cos x}\,d\cos x$

$= -\frac{4}{3}(\cos x)^{\frac{3}{2}}\Big|_0^{\frac{\pi}{2}} = \frac{4}{3}$.

12. $\int_0^{\pi/2} \sin^{10}x\,dx = \frac{10!!}{11!!} = 2\times4\times6\times8\times10 \Big/ 3\times5\times7\times9\times11$.

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No.

Date

13. $\int_0^{\pi} \sin^6 \frac{x}{2}\,dx \overset{y=\frac{x}{2}}{=\!=} 2\int_0^{\frac{\pi}{2}} \sin^6 y\,dy = 2\times\frac{5!!}{6!!}\times\frac{\pi}{2} = \frac{5\times 3}{6\times 4\times 2}\pi = \frac{5}{16}\pi.$

14. $\int_0^{\pi} (x\sin x)^2\,dx. = \int_0^{\pi} x^2\,\frac{1-\cos 2x}{2}\,dx = \underbrace{\frac{x^3}{6}\Big|_0^{\pi}}_{\frac{1}{6}\pi^3.} - \frac{1}{2}\int_0^{\pi} x^2\cos 2x\,dx$

$\int_0^{\pi} x^2\cos 2x\,dx = \frac{1}{8}\int_0^{2\pi} y^2\cos y\,dy \quad (y=2x)$

$\int y^2\cos y\,dy = \int y^2\,d\sin y = y^2\sin y - 2\int y\sin y\,dy$

$= y^2\sin y + 2\int y\,d\cos y = y^2\sin y + 2y\cos y - 2\int \cos y\,dy$

$= y^2\sin y + 2y\cos y - 2\sin y$

故 $\int_0^{2\pi} y^2\cos y\,dy = 0 + 4\pi - 0 = 4\pi$.

$\Rightarrow$ 原式 $= \frac{\pi^3}{6} - \frac{\pi}{4}$.

17. $\int_0^{\pi} \ln(x+\sqrt{x^2+a^2})\,dx$

$= \underbrace{x\ln(x+\sqrt{x^2+a^2})\Big|_0^{\pi}}_{\pi\ln(\pi+\sqrt{\pi^2+a^2})} - \int_0^{\pi} \frac{x}{\sqrt{x^2+a^2}}\,dx$

$\int_0^{\pi} \frac{x}{\sqrt{x^2+a^2}}\,dx = \frac{1}{2}\int_0^{\pi} \frac{dx^2}{\sqrt{x^2+a^2}} = \sqrt{x^2+a^2}\Big|_0^{\pi} = \sqrt{\pi^2+a^2} - |a|.$

原式 $= \pi\ln(\pi+\sqrt{\pi^2+a^2}) + |a| - \sqrt{\pi^2+a^2}$.

20. 证明：换元 $y = 1-x$

$\int_0^1 x^m(1-x)^n\,dx = \int_1^0 (1-y)^m y^n(-1)\,dy = \int_0^1 y^n(1-y)^m\,dy$

$= \int_0^1 x^n(1-x)^m\,dx.$ □

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